Universidad de los Andes

Cálculo Integral con Ecuaciones (202319)

Ejercicios para practicar

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11.6 Test del radio y de la raiz

(1) Determine si la serie converge condicionalmente, converge absolutamente o diverge.

(a) $\displaystyle\sum_{n=0}^{\infty}(-1)^n\frac{n}{2n^3+1}$

(b) $\displaystyle\sum_{n=1}^{\infty}\frac{(-1)^{n+1}}{\sqrt[4]{n}}$

(c) $\displaystyle\sum_{n=1}^{\infty}\frac{10^n}{(n+1)4^{2n+1}}$

(d) $\displaystyle\sum_{n=1}^{\infty}\frac{(-1)^{n+1}n^2 2^n}{n!}$

(e) $\displaystyle\sum_{n=1}^{\infty}\frac{(-1)^n\arctan(n)}{n^2}$

(f) $\displaystyle\sum_{n=1}^{\infty}\left(\frac{n^2+1}{2n^2+1}\right)$

(g) $\displaystyle\sum_{n=1}^{\infty}\frac{n!}{n^n}$

(h) $\displaystyle\sum_{n=1}^{\infty}\left(\frac{-2n}{n+1}\right)^{5n}$

(i) $\displaystyle\sum_{n=1}^{\infty}\left(1+\frac{1}{n}\right)^{n^2}$

(j) $\displaystyle\sum_{n=1}^{\infty}\frac{3-\cos n}{n^{2/3}-2}$